Baseline Results

Dual-View Address Mapping for NPU–PIM Memory on Ramulator 2 — Group 13

Yash Gupta, Nikunj Ladumor, Vamshi Kumar Kanduri, Priyanshu Ganjeer — Dept. of CSE, IIT Guwahati

Setup. Two fixed address layouts for the 15 line-index bits of a 2 MB page, listed from bit 0 upward: **NPU-friendly** (bank group, column, bank, row) and **PIM-friendly** (column, bank group, bank, row). NPU utilization is measured in Ramulator 2.1 (DDR4-2400R, one channel) for a sequential read stream (utilization $= 4\times$reads/DRAM cycles). PIM utilization is an analytical all-bank GEMV model using Ramulator's DDR4 timings. Weights are the LLaMA3-8B matrices: six with d_{in}=4096 and `down_proj` with d_{in}=14336.

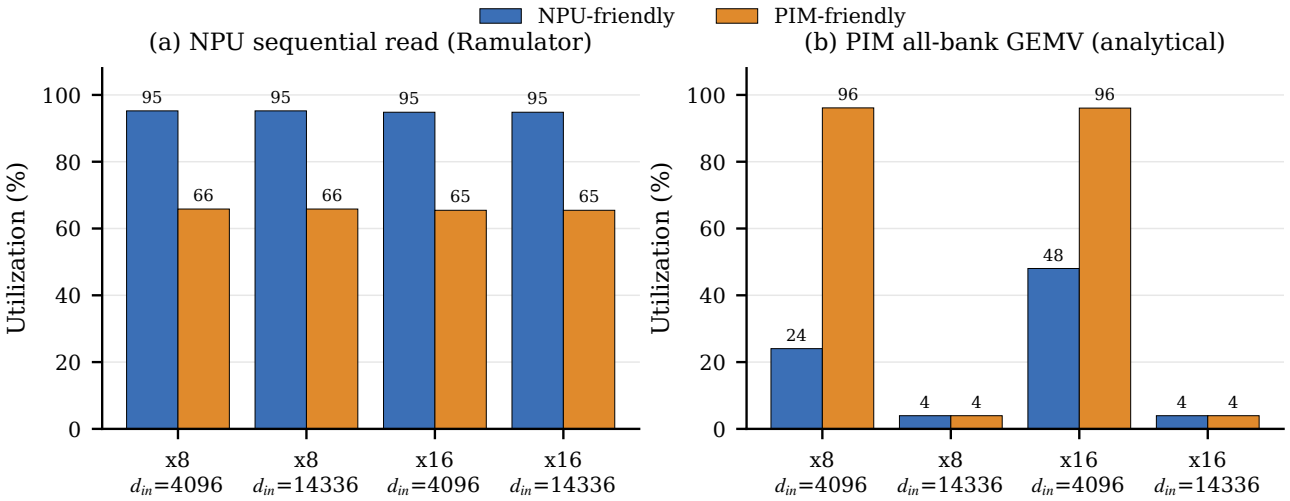


Figure 1: Baseline bandwidth utilization. (a) NPU sequential read (Ramulator); (b) PIM all-bank GEMV (analytical).

Table 1: Baseline utilization (%) per case.

Case	Layout	NPU	PIM
x8, d_{in}=4096	NPU-friendly	95.23	24.03
x8, d_{in}=4096	PIM-friendly	65.83	96.12
x8, d_{in}=14336	NPU-friendly	95.23	3.95
x8, d_{in}=14336	PIM-friendly	65.83	3.95
x16, d_{in}=4096	NPU-friendly	94.82	48.03
x16, d_{in}=4096	PIM-friendly	65.46	96.06
x16, d_{in}=14336	NPU-friendly	94.82	3.95
x16, d_{in}=14336	PIM-friendly	65.46	3.95

Table 2: Baseline utilization (%) over all seven weight matrices of a decoder layer, weighted by their size in 2 MB pages of weight data (152 pages with d_{in}=4096, 56 for `down_proj`).

DRAM	Layout	NPU	PIM
x8	NPU-friendly	95.23	18.62
x8	PIM-friendly	65.83	71.30
x16	NPU-friendly	94.82	36.16
x16	PIM-friendly	65.46	71.26

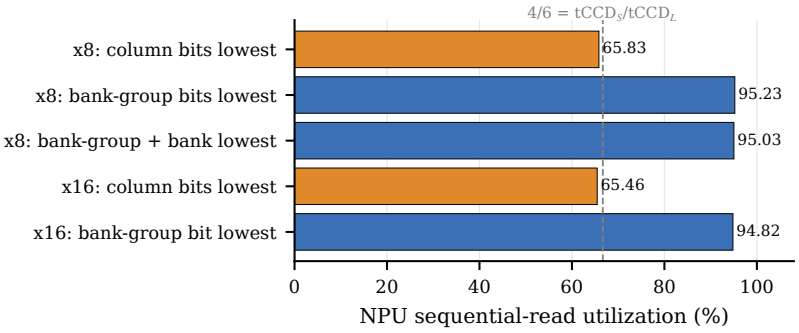


Figure 2: Why the NPU-friendly layout is fast: with a bank-group bit lowest, consecutive reads alternate bank groups (4-cycle spacing, t_{CCD_S}); with column bits lowest every read waits 6 cycles (t_{CCD_L}), capping utilization near $4/6 = 66.7\%$.

Observations.

- Each fixed layout serves one device only: NPU-friendly gives about 95% NPU but 24% PIM (x8, d_{in}=4096), because two of the four bits that select a PIM unit are column bits and four units share each bank. PIM-friendly gives 96% PIM but only about 66% NPU.
- For `down_proj` both layouts give 3.95% PIM: its unit-selection bits (9–12) contain two row bits, so units share banks and switch rows on every access. Both baselines are the standard orders and are designed for d_{in}=4096.